# Technical specification template: Power Transformers

*OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A6. Generated 2026-11-08 12:00 UTC. Package version 0.1.0.dev0.*

**DRAFT TEMPLATE. NOT AN ISSUED SPECIFICATION. NOT FOR PROCUREMENT. PRELIMINARY VALUES ARE NOT MANUFACTURER-CONFIRMED.**

## Document control

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## 1. Scope

This template defines the technical requirements for power transformers for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

## 2. Applicable standards

- IEC 60076 series (power transformers)
- IEC 60076-7 (loading guide, mineral-oil-immersed)
- IEC 60076-11 (dry-type), if dry-type is selected
- IEC 60076-5 (ability to withstand short circuit)
- Local utility and authority requirements [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

## 3. Service conditions and system data

| Item | Requirement or preliminary value | Data status |
|---|---|---|
| Nominal LV system | 400 V line-to-line, 60 Hz, 3-phase | PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified |
| MV supply | 13.8 kV | PRELIMINARY_ASSUMPTION. To be confirmed with the utility |
| Design ambient temperature | 30 degC | PRELIMINARY_ASSUMPTION. Site data to be confirmed |
| Site altitude | 0 m | PRELIMINARY_ASSUMPTION. Site data to be confirmed |
| Prospective fault level | [To be calculated: short-circuit | NOT CALCULATED |

| | | |
|---|---|---|
| | study pending utility, transformer and generator data] | |

## 4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

| Item | Requirement or preliminary value | Data status and note |
|---|---|---|
| Quantity (total) | 2 | PRELIMINARY_ASSUMPTION |
| Rated power (each) | 3,150 kVA | UNVERIFIED_PLACEHOLDER. Loading limits are applied to the derated rating |
| Short-circuit impedance | 6 % | UNVERIFIED_PLACEHOLDER. Guaranteed value and tolerance from the manufacturer |
| Primary / secondary voltage | 13.8 kV / 400 V | PRELIMINARY_ASSUMPTION |
| Type and cooling | [Dry-type or liquid-immersed; cooling class: project to select] | Project decision |
| Vector group, tap changer | [Project to select] | Project decision |
| Derating (ambient, altitude, installation) | 1.0 | UNVERIFIED_PLACEHOLDER. 1.00 is a placeholder. Manufacturer's loading guide governs |

## 5. Technical requirements

5.1 The transformer shall carry its rated power continuously at the site conditions in the service-conditions table without exceeding the temperature-rise limits of the applicable standard.

5.2 The manufacturer shall state the continuous rating at the site conditions and any derating for ambient temperature, altitude and enclosure.

5.3 The manufacturer shall not rely on short-duration emergency overload to meet the continuous loading requirement.

5.4 The short-circuit impedance, tolerance, no-load and load losses shall be guaranteed values supplied in the tender and confirmed by test.

5.5 The transformer shall withstand the thermal and dynamic effects of the specified through-fault level for the specified duration; evidence by test or calculation per the applicable standard.

5.6 Losses evaluation basis: [project to define].

5.7 Noise level, enclosure protection rating, temperature monitoring and protection accessories: [project to define].

5.8 Number of transformers per power path: 1; each path is to carry the whole facility load after loss of the other path.

## 6. Vendor data schedule

The tenderer shall complete the schedule below with its offer: the value offered, its unit, and the name and revision of the document that states it. A value without a source document is not accepted. The parameters are those in the manufacturer-data register; supplied values are imported into the calculations through the data override mechanism, and a value stays NOT VERIFIED until a reviewer records an attestation against its source (name, organisation, date, evidence reference, scope).

| Parameter id | Parameter | Datasheet required | Current status | Value offered | Unit | Document and revision |
|---|---|---|---|---|---|---|
| equipment.transformer_rating | Transformer rating | Transformer data sheet | UNVERIFIED_PLACEHOLDER | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| equipment.transformer_impedance | Transformer impedance | Transformer data sheet | UNVERIFIED_PLACEHOLDER | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| losses.distribution_loss_fraction | Distribution losses | Transformer loss report and busduct and cable loss data | UNVERIFIED_PLACEHOLDER | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| derating.transformer_ambient_factor | Transformer ambient derating factor | Manufacturer derating curve for ambient conditions (transformer) | UNVERIFIED_PLACEHOLDER | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| derating.transformer_altitude_factor | Transformer altitude derating factor | Manufacturer derating curve for altitude conditions (transformer) | UNVERIFIED_PLACEHOLDER | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| derating.transformer_installation_factor | Transformer installation derating factor | Manufacturer derating curve for installation conditions (transformer) | UNVERIFIED_PLACEHOLDER | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| derating.transformer_contingency_factor | Transformer contingency derating factor | Transformer data sheet | UNVERIFIED_PLACEHOLDER | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| short_circuit.utility_fault_level_mva | Utility initial short-circuit power at the MV connection | Utility written statement of maximum and minimum fault levels and X/R at the point of connection | MISSING | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| short_circuit.transformer | Transformer X/R ratio | Transformer test report | MISSING | [vendor to complete] | [vendor to complete] | [vendor to complete] |

| | | | | | | |
|---|---|---|---|---|---|---|
| _x_r_ratio | | (short-circuit impedance and load losses) | | | | |

Model, manufacturer, country of manufacture, and the type-test reports for the offered design: [vendor to complete].

## 7. Compliance schedule

The tenderer shall state, for every standard in section 2 and every technical requirement in section 5, whether the offer complies. 'Complies' needs an evidence reference (type-test report, certificate, calculation, drawing or data sheet with revision). A statement without evidence is recorded as not demonstrated. The requirement references below are the numbered items of section 5 of this template; no clause of any standard is quoted.

| Standard named in section 2 | Edition offered | Complies / complies with comment / does not comply | Evidence reference | Comment or deviation |
|---|---|---|---|---|
| IEC 60076 series (power transformers) | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| IEC 60076-7 (loading guide, mineral-oil-immersed) | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| IEC 60076-11 (dry-type), if dry-type is selected | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| IEC 60076-5 (ability to withstand short circuit) | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| Local requirements [project to add] | [vendor to complete] | [vendor to complete] | [vendor to complete] | [vendor to complete] |

| Requirement (section 5) | Complies / complies with comment / does not comply | Evidence reference | Comment or deviation |
|---|---|---|---|
| 5.1 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.2 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.3 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.4 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.5 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.6 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.7 | [vendor to complete] | [vendor to complete] | [vendor to complete] |
| 5.8 | [vendor to complete] | [vendor to complete] | [vendor to complete] |

## 8. Inspection and quality requirements

8.1 Review of design calculations, drawings and the inspection and test plan before manufacture.

8.2 Inspection of materials certificates and core and winding processing at hold points defined in the ITP [project to define].

8.3 Dimensional and nameplate check before shipment.

8.4 Routine tests on every unit as required by the applicable standard and this specification.

8.5 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

8.6 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

8.7 Calibration certificates for test instruments.

## 9. Factory acceptance test (FAT) requirements

9.1 Routine tests per the standard: ratio and vector group, winding resistance, no-load loss and current, load loss and impedance voltage, applied voltage, induced voltage, insulation resistance.

9.2 Temperature-rise test or evidence on a duplicate unit [project to define].

9.3 Witnessing: [owner / independent inspector to attend]. Results recorded against the guaranteed values.

9.4 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9.5 Acceptance criteria are taken from the applicable standard and this specification [project to define]; a test passed against a criterion not stated in advance is not accepted.

9.6 Equipment shall not be shipped until the FAT report is accepted or a concession is recorded and approved by the owner.

## 10. Site acceptance test (SAT) requirements

10.1 Inspection for transport damage, nameplate and accessories; insulation resistance and ratio on site; confirmation of tap position, protection settings and alarm functions.

10.2 A SAT procedure shall be submitted for approval before the tests. It states the test, the method, the instrument, the acceptance criterion and the witness, with the criteria taken from the applicable standard and this specification [project to define].

10.3 SAT is carried out on the installed equipment with its final connections, after the installation inspection and before commissioning loads are applied.

10.4 Results are recorded on signed sheets. Any deviation, retest and concession is listed with its disposition. A SAT is not passed while a deviation is open.

10.5 SAT results are compared with the FAT results; a significant difference is investigated and recorded.

## 11. Commissioning requirements

11.1 Visual and mechanical inspection, torque checks, insulation resistance, ratio and polarity.

11.2 Protection and temperature-alarm function tests.

11.3 Energisation sequence and thermography under load [project to define].

11.4 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

11.5 Integrated systems testing of the whole power chain (utility loss, generator start and transfer, UPS on battery, loss of one power path, return to normal) is carried out under a project procedure [project to define]. This equipment's commissioning is not complete until it has taken part in that test.

## 12. Submittals and documentation

| Document | When required | Status |
|---|---|---|
| Completed vendor data schedule (section 6) and compliance schedule (section 7) | With the tender | [vendor to complete] |
| Datasheets and general arrangement drawings | With the tender, final before manufacture | [vendor to complete] |
| Type-test reports and certificates for the offered design | With the tender | [vendor to complete] |
| Inspection and test plan (ITP) | Before manufacture | [vendor to complete] |
| FAT procedure and report | Before and after the FAT | [vendor to complete] |
| Installation instructions and SAT procedure | Before delivery | [vendor to complete] |
| Operation and maintenance manuals, as-built drawings and final data | Before commissioning; as-built at handover | [vendor to complete] |
| Recommended spare parts and special tools lists | With the tender and at handover | [vendor to complete] |
| Training plan and materials [project to define] | Before commissioning | [vendor to complete] |
| Warranty terms [project to define] | With the tender | [vendor to complete] |

## 13. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- The vendor data schedule (section 6) and the compliance schedule (section 7) are blank until a tenderer completes them.
- FAT, SAT and commissioning acceptance criteria are [project to define].
- This template has not been reviewed by an independent engineer.